Improving Kitchen Prep Time (KPT)

Beyond Timestamps:
Predicting KPT using Kitchen Physics & Merchant Integrity

Zomathon 2026

Problem Statement-1

Team: The Outliers

Team members:
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The Problem

- Current KPT predictions rely heavily on merchant-marked “Food Ready” (FOR) signals.
- In practice, these signals are shaped more by human behavior, kitchen pressure, and incentives than by the actual moment food is ready.

Root Causes of KPT Signal Distortion

Rider-Triggered FOR Events — **01**

“Food Ready” is often marked when the rider arrives, not when preparation is complete.

Unobserved Kitchen Load — **02**

The system only observes Zomato orders, while in-store dining and competitor orders also contribute to kitchen load

Human and operational bias — **03**

Manual marking behavior varies significantly across merchants and time periods

Downstream Operational Impact

Riders — **01**

Wrong KPT means riders often reach too early and end up waiting. This cuts into their earnings, lowers motivation, and increases churn.

Customers — **02**

When KPT is off, ETAs keep changing.
This “ETA yo-yo” makes delivery feel unreliable and hurts customer trust.

Platforms — **03**

Noisy KPT breaks batching and dispatch decisions, leading to poor rider utilization and higher cost per delivery.

Model-only solutions are insufficient. Signal quality must be improved.

18-Signal Architecture & Impact

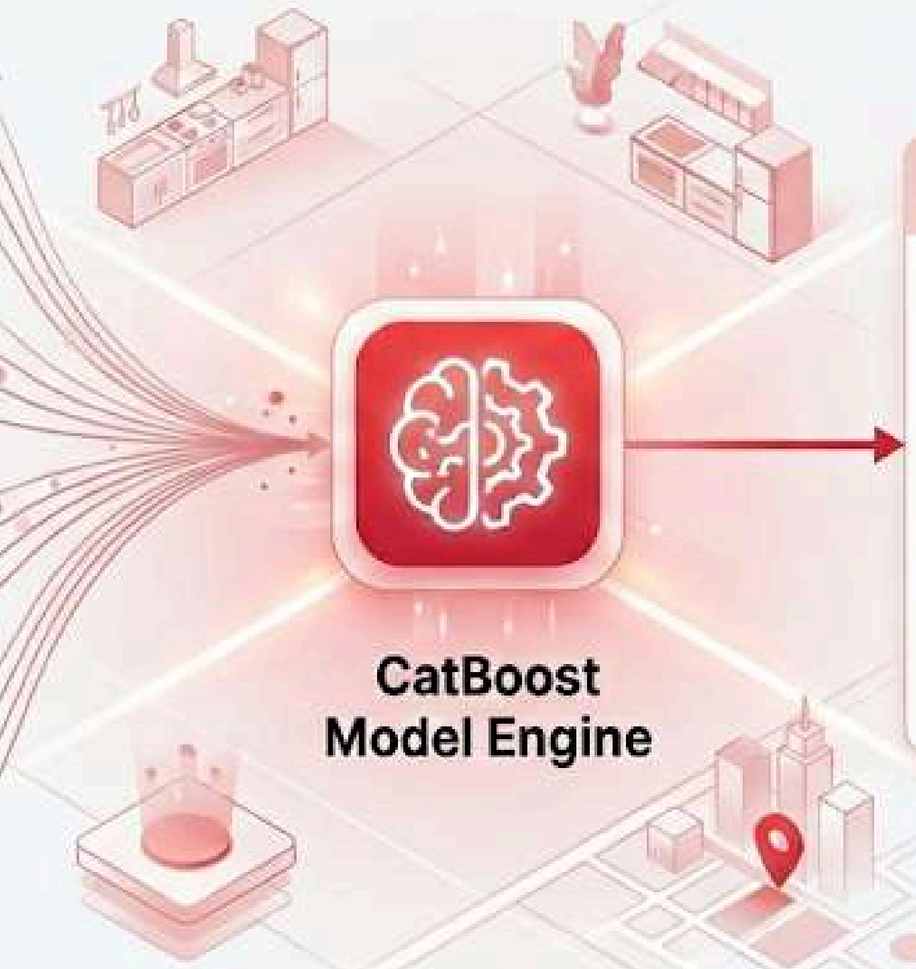
A four-layer data science architecture for kitchen preparation time (KPT) prediction)

Layer 1: Internal Load
- Total Item Quantity
- Pending Queue
- Items in Order

Layer 2: Chef Psychology
- Context-Switch Penalty
- Instruction Friction
- Packaging Complexity

Layer 3: Environmental Proxies
- Dine-in Crowd Proxy
- Weather Impact
- Vertical Transit Cost

Layer 4: Historical Anchors
- Past KPT Baseline
- Store ID
- Day of the Week
- Rating



Operational Improvement

↓ 0.41 Min
MAE Reduction

↑ 685
Rider Hours Saved/Day

11.57% improvement

Enhanced Signals Used to Improve KPT Prediction

Layer-1

The "Hidden" Stress Signal

- **WiFi Crowd Proxy**
- **Rush Window Bucket**
- **Ending Queue**

01 Uses signal strength to "see" dine-in customers Zomato misses

02 Live ratio of order velocity vs. kitchen bandwidth

03 Real-time backlog tracking of the live kitchen stack

Expose real kitchen pressure by capturing demand and congestion invisible to merchant-marked signals.

Enhanced Signals Used to Improve KPT Prediction

Layer-2

The "Chef Psychology" Signals

- **Context-Switch Penalty**
- **Instruction Friction**
- **Packaging Complexity**

Model human cognitive and workflow friction inside the kitchen that directly slows preparation.

04 Measures the cognitive delay when a chef switches between diverse dish type

05 NLP-derived index for "Customization" keywords that break kitchen rhythm

06 Scoring based on dish state (Liquids/Gravies require +X mins for sealing)

Enhanced Signals Used to Improve KPT Prediction

Layer-3

Geospatial & Atmospheric Signals

Vertical Transit Cost

07 Predicts the "Mall Floor Delay"- time spent in elevators/stair

Weather Impact

08 Humidity and Rain intensity factored into frying times and hand-off friction

Account for physical and environmental constraints that delay readiness beyond cooking itself.

Enhanced Signals Used to Improve KPT Prediction

Layer-4

The Anchor Signal

Past KPT Baseline

09 30-day "Normal" anchor for specific store/time slots

Merchant Reputation

10 A de-biasing score to penalize "Early-Button Clickers"

Stabilize predictions using trusted historical norms while correcting systematic merchant bias.

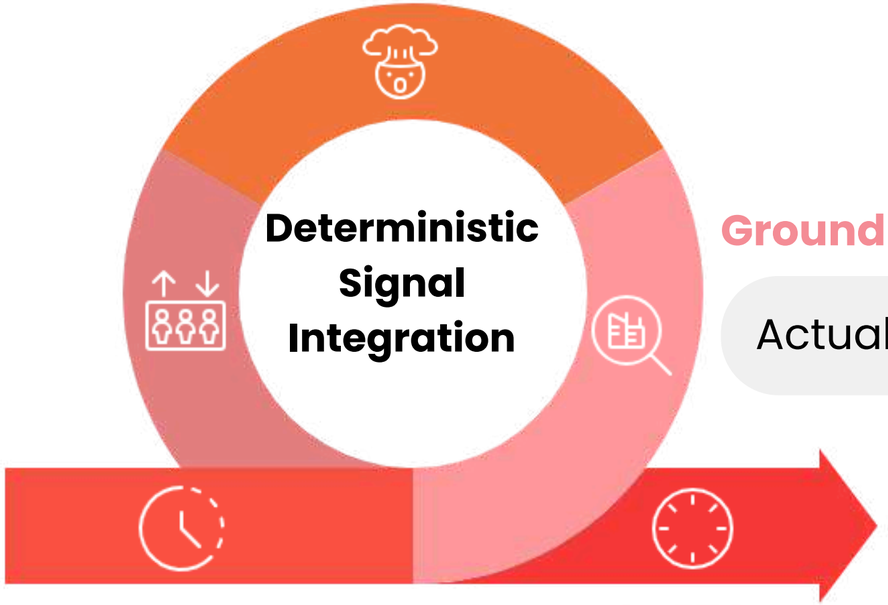
De-Noising the "Truth": From Human Bias to Deterministic Signals

Invisible Stress Signals

- **Dine-in Crowd Proxy**: Captures hidden dine-in load missed by Zomato systems, reducing peak-hour ETA spikes.
- **Context-Switch Penalty & Instruction Friction:** Measures how difficult items are to prepare together, not just how many.

Environmental Drag

- **Vertical Transit Cost:** De-noises KPT by separating cooking time from predictable mall and floor-level delays.
- **Kitchen Efficiency Momentum:** Adapts KPT in real time by detecting when kitchens accelerate or approach overload.



Ground Truth Training

Actual KPT Duration

Inaccurate ETAs

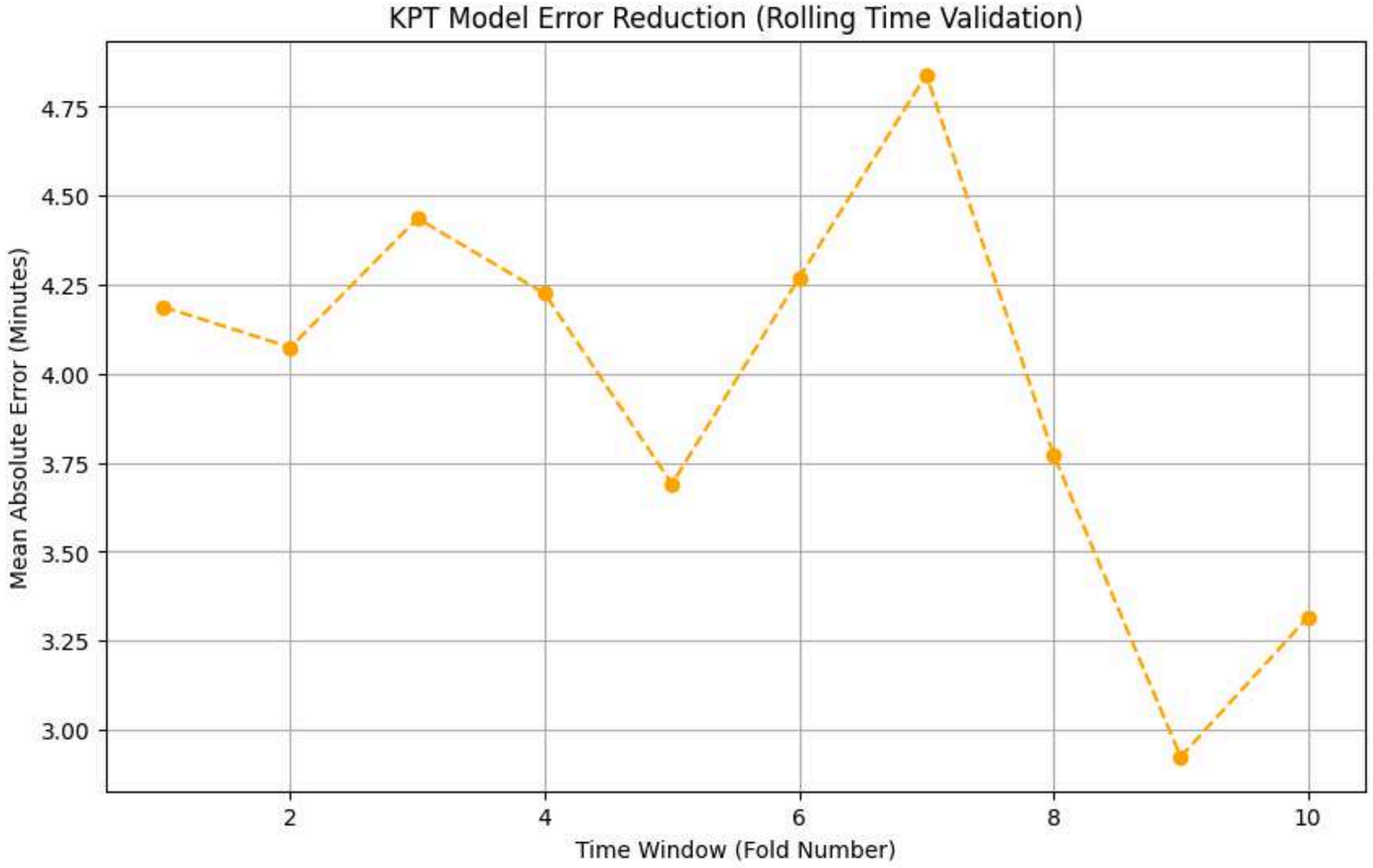
Human bias skews predictions

Accurate ETAs

Deterministic signals improve predictions

Rolling Window Cross-Validation:

- We used a rolling window to show how the **model improves** as it sees more data.
- **Learning Over Time**: Our signals help the model "learn" the kitchen's rhythm, which steadily brings the MAE (error) down as the training window grows.



Proprietary Dataset

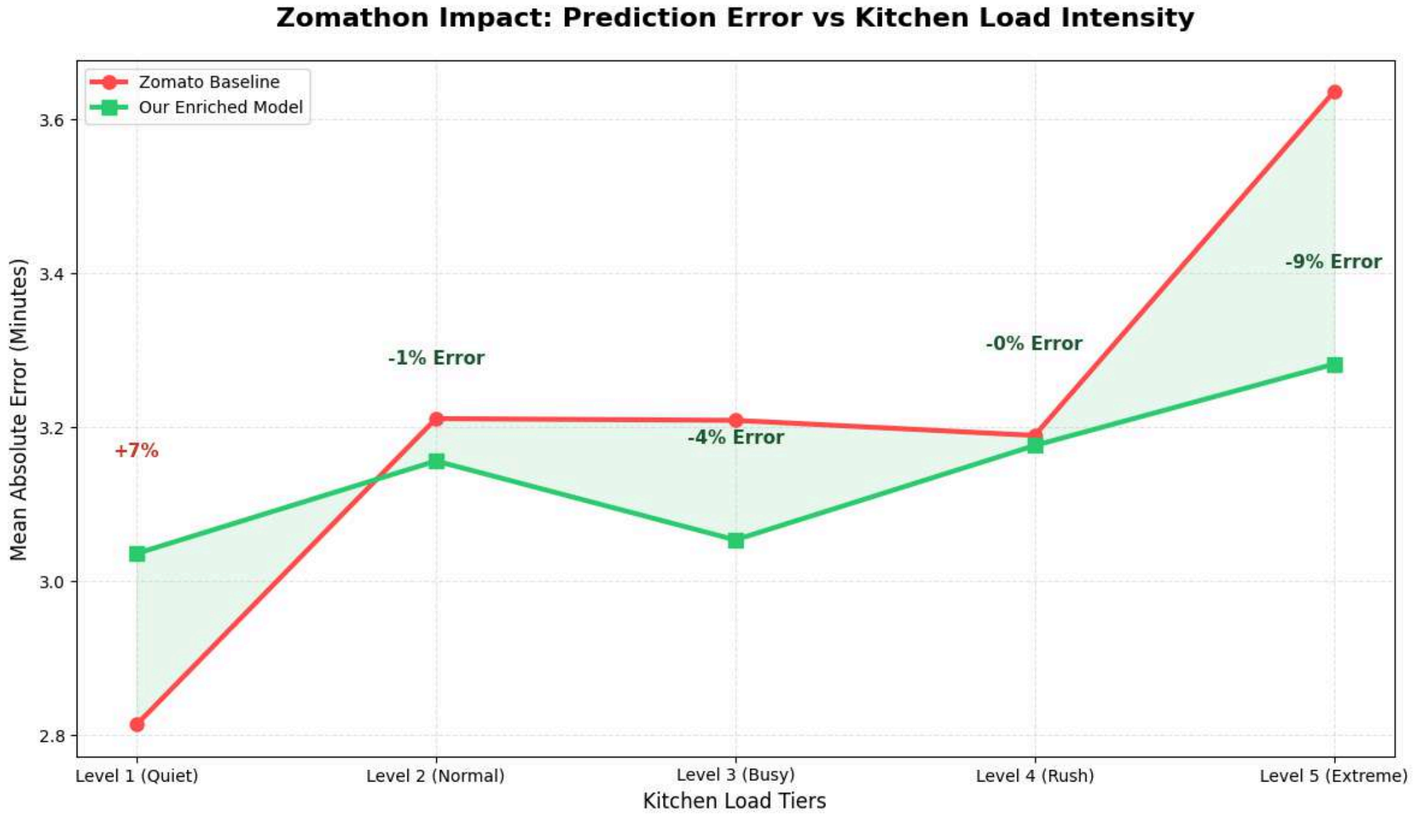
Scale: 31,000+ Unique Orders (Self-curated for high variance).
Feature Depth: 18 Custom Signals (Environmental + Psychological + Physical).
Data Integrity: Chronological sorting to ensure Zero Data Leakage.

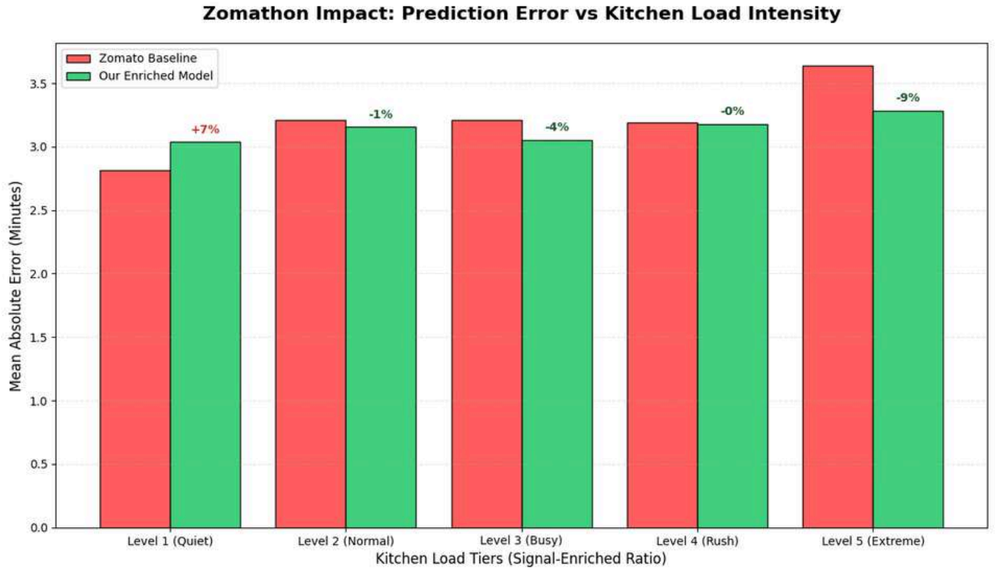
Improvements in ETA at diff kitchen stress levels

- **Performance Under Pressure**: Our model shows the biggest improvement during high kitchen stress levels.
- **Why the Baseline Fails**: Standard models usually lose accuracy during a rush because they lack real-time signals.
- **The Solution**: By including features like WiFi sensing and the context-switch penalty, we significantly reduce prediction errors even when the kitchen is at its limit.

- **9%** reduction in error in high kitchen load
- **4%** reduction in moderate load

(-ve error means our model is performing nicely)





The Core Metric: Rider Efficiency

A seemingly small improvement → just **0.41 minutes per order** → compounds into a massive efficiency win at scale.

At 100,000 orders per day, this translates to:

- ~685 rider hours saved every single day
- 250,000+ hours saved annually
- Hours that move riders from waiting outside kitchens to earning on the road

$$\text{Rider Hours Saved} = \left(\frac{\text{Total Orders} \times \text{Reduction in MAE per order}}{60} \right)$$

685.07 hrs/day saved

Bottom Line Impact:

01 Rider Churn
Higher hourly earnings through reduced frustration.

02 Refund Rates
Better thermal synchronization for perishables like ice cream.

03 Order Batching Efficiency
Reliable KPTs allow Zomato to pair orders with 100% confidence.

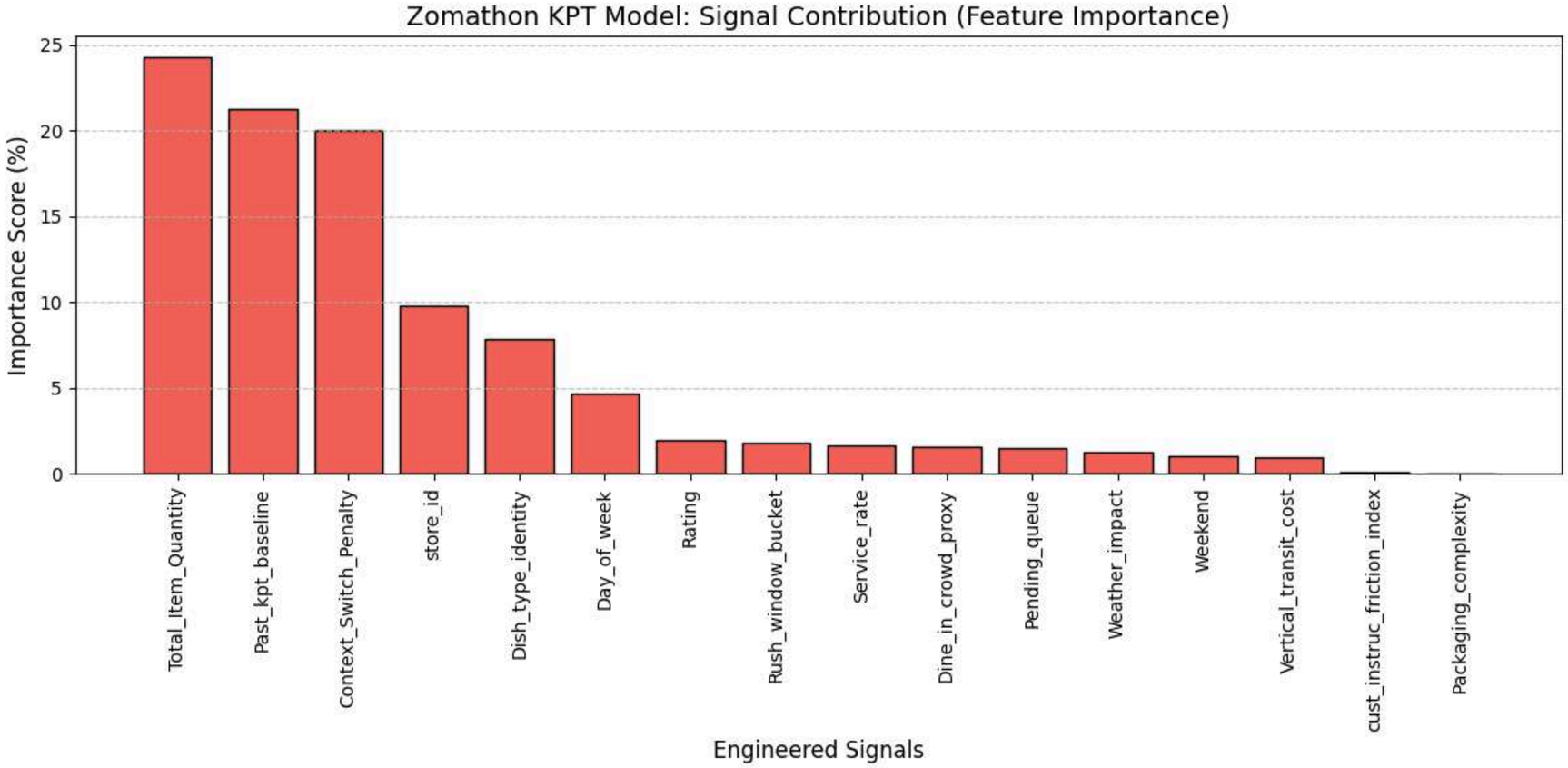
Signals importance in the model

The "Core Logic" (Top 65% Impact)

- **Total Item Quantity**: The primary driver of physical labor time.
- **Past KPT Baseline**: The "Historical Anchor" that stabilizes predictions against store-specific norms.
- **Context Switch Penalty:** Our key innovation that quantifies the cognitive delay when a chef manages diverse dish types.

The "Stress & Friction" Layer

- **Store & Dish Identity:** Captures unique operational signatures and category-specific prep speeds.
- **Dine-in Crowd & Pending Queue**: Solves the "Invisible Load" problem by factoring in offline customers and real-time backlogs.
- **Vertical Transit & Weather**: Normalizes for environmental "drag" like mall floor delays and rain-related handover friction.



Our Impact

Rider

Earning with Dignity

- **The Reality**: Riders lose income standing at counters due to "Fake Ready" signals.
- **The Human Win**: We save **685.07** hours of idle time daily (at 100k scale).
- **The Result**: We transform "unpaid waiting" into "earning time," respecting the rider's hustle and fuel.

Customer

Reliability You Can Trust

- **The Reality**: "ETA Yo-Yos" and melted food destroy the dinner experience.
- **The Human Win**: **0.41-minute precision** gain per order through thermal and physical de-noising.
- **The Result**: Parents can time dinner and workers can plan breaks, ensuring food arrives exactly when expected.

Merchant

Calm in the Kitchen

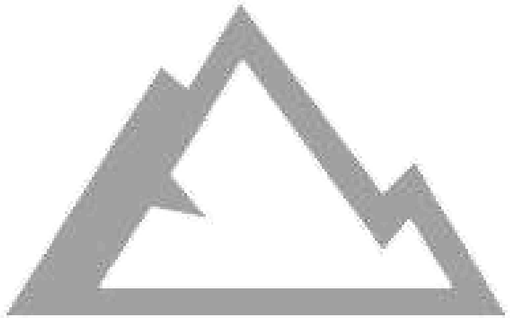
- **The Reality**: "Invisible" dine-in crowds and rider pressure cause chef burnout.
- **The Human Win**: Capturing the "**Context-Switch Penalty**" and "**Dine-in Load**" via our new signal layers.
- **The Result**: A synchronized flow that lets chefs focus on their work rather than managing crowd.

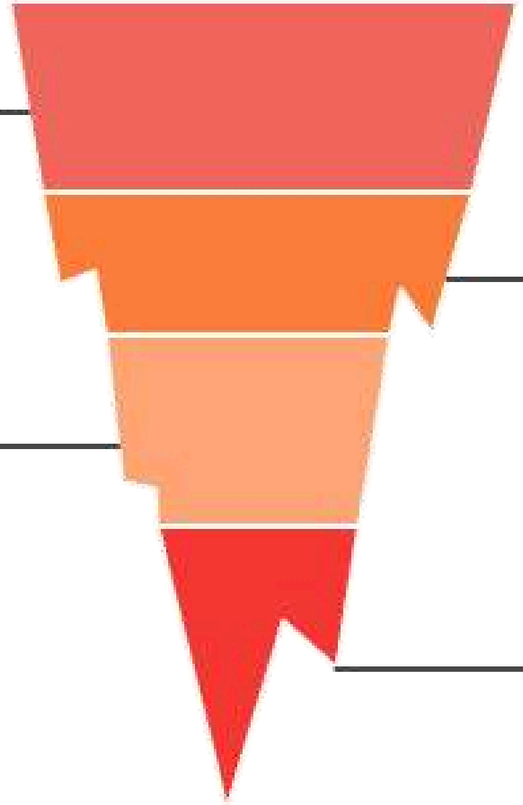
We aren't just optimizing a variable' we are optimizing the lives of those who power the platform. Every 0.41 minutes saved is a victory for the person behind the bag and the person at the door

Our Impact

Unpaid Rider Hours

Riders waste time at counters.





Human Impact on Riders

Model saves 685 hours of idle time.

Merchant Kitchen Chaos

Pressure from riders causes kitchen errors.

Customer Reliability

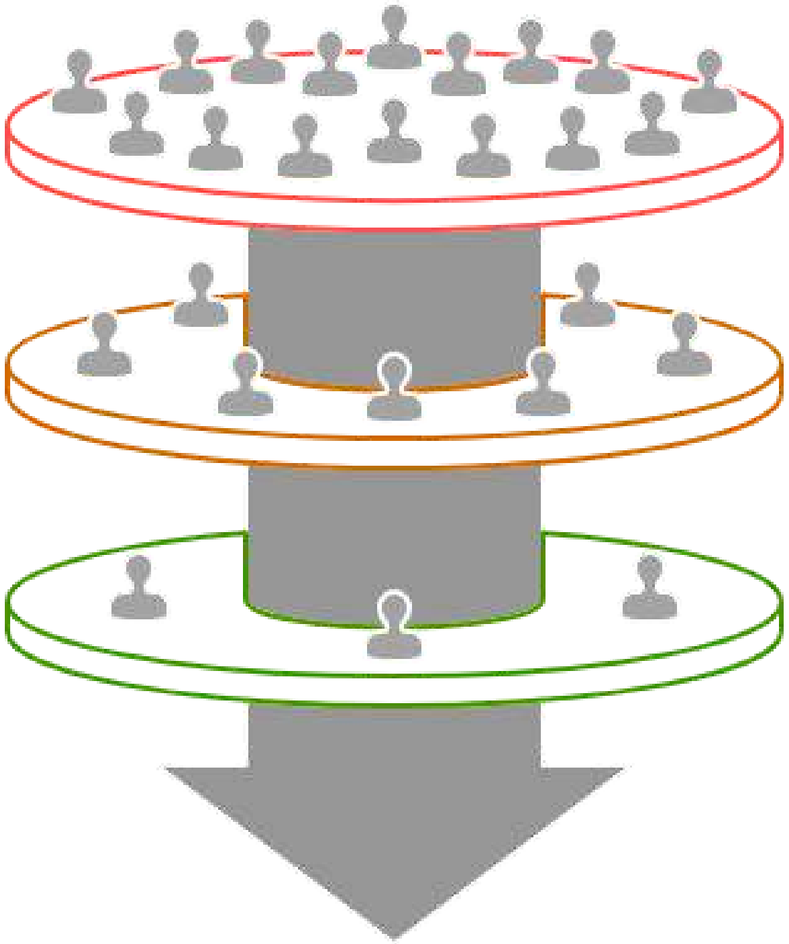
0.41-minute gain in ETA precision.

De-noising the Invisible Load

Dine-in Crowd Proxies reduce kitchen pressure.

Tiered Implementation Strategy

We recognize that a "one-size-fits-all" approach fails at Zomato's scale. Our deployment is categorized by merchant volume.



Tier 1 (High-Volume / Malls): Full signal integration — WiFi/Bluetooth crowd sensing + vertical transit cost modeling to handle complex logistics.

Tier 2 (Mid-Market): Leverage merchant app data to model chef context-switching and packaging complexity.

Tier 3 (Small-Scale / Street-side): Use historical KPT baselines and weather impact for accurate ETAs — no extra hardware needed.

Technical Scalability & Global Impact

Scalable Infrastructure

- **CatBoost Engine**:
  High-speed inference across 100K+ concurrent orders
- **Edge-Ready NLP**:
  Async processing (Instruction Friction Index) , zero Mx app lag
- **Privacy-First Signals:**
  Anonymized dine-in density captures invisible load safely

Platform-Level Impact

- **Self-Correcting Signals**:
  De-biased “Ready” input improves with multi-city data
- **Operational ROI**:
  0.41 min MAE ↓ → ~685 rider hours saved daily
- **Future-Proof Architecture**:
  Plug-and-play IoT integration (e.g., kitchen thermal sensors)

Our architecture is built for the street-side stall and the 5-story mall alike—ensuring every rider's time is respected

Links to resources:

01 Github: https://github.com/yashtayalcse/Zomathon-Project

02 Dataset: Self-curated-KPT-dataset-link

03 NoteBook: kitchen-prep-time-dataset-preprocessing.ipynb